

Beyond Language Barriers

Evaluating the Efficacy of general-purpose large language models in Analyzing Sentiment in French Textual Data

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Abstract

This study examines whether general-purpose large language models (LLMs) can effectively analyze sentiment in non-English texts without task-specific fine-tuning. We evaluate 11 open and closed source models on 200 sentences randomly drawn from a corpus of 2,683 French news articles discussing open-source software, comparing their performance to traditional dictionary methods. Through 19,800 systematically designed prompts across three language conditions (French prompts with French text, English prompts with French text, and English prompts with English translations), we find that leading closed source models achieve strong sentiment analysis performance ($r > 0.70$). Open source models demonstrate viable capabilities correlating with parameter size. We observe only marginal benefits from cross-linguistic prompting. All models perform substantially better on three-category classification (positive/neutral/negative) than on more granular sentiment distinctions. These findings suggest that general-purpose LLMs offer accessible alternatives to specialized sentiment models for social science researchers, particularly when basic sentiment polarity detection is sufficient.

Keywords: Sentiment Analysis, Large Language models, Cross-lingual NLP, French Language, Open-Source models, Computational Social Science, Prompt Engineering

Introduction

The rise of large language models (LLMs) has transformed natural language processing across domains and applications. While these models demonstrate impressive capabilities in non-English languages, their cross-linguistic effectiveness remains a critical area of investigation. Indeed, the majority of foundational LLMs, such as Meta AI's Llama models, are predominantly trained on English data (?), raising questions about their performance in other languages.

This paper addresses a fundamental question in this domain: Can open and closed-source general-purpose foundational LLMs accurately evaluate sentiment in non-English texts?²

Recent research has uncovered a compelling phenomenon: prompting in English often improves performance in cross-linguistic tasks, even when the target language is non-English. ? documented increased accuracy in knowledge extraction when using English prompts compared to diversified manual prompts in other

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²While 'open source' traditionally implies access aligned with FOSS principles, its usage within the LLM context often refers more specifically to the public release of model weights, a condition more accurately described as 'open weights'. This paper adopts this common LLM terminology, using 'open source' primarily to denote models providing greater accessibility through available weights, distinct from closed-source models. We acknowledge this usage differs from stricter FOSS definitions, and that most models in our 'open source' category are technically 'open weights'.

languages. This finding was corroborated by ?, who observed that English prompts outperform multilingual configurations across natural language understanding and machine translation tasks in 30 languages.

This pattern extends beyond isolated studies. ?, ?, have all reported significant performance gains—measured by accuracy, F1 scores, and other metrics when models are guided with English instructions, regardless of the target language. These findings suggest a fundamental characteristic of current LLM architectures that merits deeper investigation.

While specialized models like BERT and its variants have been fine-tuned specifically for sentiment analysis tasks and perform effectively, they often require technical expertise to implement and adapt. By contrast, using non-fine-tuned foundational models could provide a more accessible solution for non-technical social science researchers who may lack the computational resources or programming knowledge necessary for model customization. This accessibility consideration is particularly relevant as AI tools become increasingly integrated into research methodologies across disciplines.

Research Question

Sentiment analysis serves as an ideal test case for examining the cross-linguistic capabilities of LLMs for several reasons. First, it requires nuanced language understanding beyond simple translation, demanding models grasp cultural contexts and linguistic subtleties. Second, it represents a common analytical task with practical applications across multiple disciplines. Third, sentiment analysis offers both continuous (intensity) and categorical (polarity) evaluation metrics, enabling multifaceted performance assessment. Finally, the availability of established dictionary-based methods provides meaningful benchmarks against which to evaluate LLM performance, creating a comprehensive evaluation framework for our research questions.

Thus, the main goal of this paper is to answer the following questions.

1. Can general-purpose LLMs accurately evaluate sentiment in non-English texts without task-specific fine-tuning?
2. Can open-source general-purpose foundational LLMs accurately evaluate sentiment in non-English texts without task-specific fine-tuning?
3. How does the language of the prompt affect performance across different languages?

Sentiment analysis offers an interesting test case for examining this phenomenon, as it requires nuanced language understanding beyond simple translation. While closed-source models such as those offered by Google or Anthropic have received substantial attention in multilingual contexts, open-source LLMs present both unique challenges and opportunities. They offer greater transparency, adaptability, and accessibility for global researchers and practitioners, yet their cross-linguistic capabilities remain less comprehensively documented. This study first evaluates the performance of leading open and closed-source foundational LLMs in sentiment analysis with French texts. It then examines how prompt language affects performance, identifies cross-linguistic patterns, and analyzes model behavior across French and English. Our findings seek to contribute to a deeper understanding of LLM’s capabilities and limitations in multilingual contexts and offer practical guidelines for optimizing their deployment in diverse language settings.

Literature Review

Specialized Models for Multilingual Sentiment Analysis

Previous research consistently demonstrates the effectiveness of specialized models for sentiment analysis in non-English languages. These models, based on various transformer architectures, have achieved impressive performance across diverse linguistic contexts. Transformer-based architectures have shown particularly strong results, with accuracy rates often exceeding 85% across various Indo-European and Semitic languages (??). Researchers have reported up to 95% accuracy for Bengali sentiment analysis (?), while others achieved 96% accuracy for Greek sentiment analysis (?). For Arabic, ? documented 96.442% accuracy using specialized models, demonstrating the effectiveness of language-specific fine-tuning.

These language-specific adaptations consistently outperform general multilingual approaches. ? and ? developed specialized models for Arabic and Indonesian respectively, both reporting significant improvements over generic multilingual models. This pattern holds across diverse language families, with specialized models for South Asian, East Asian, and Semitic languages demonstrating strong performance within their target domains (??). Even for lower-resource languages, fine-tuned models have shown promise. ? demonstrated competitive performance for Tigrinya, while ? reported encouraging results for Tamil and Tulu using specialized transformer-based architectures. ? documented F1 scores between 62% and 83% for various African languages, noting that performance improved with increased training data availability.

The literature also shows progress in handling code-mixed and transliterated text, particularly relevant for many South Asian language contexts. ? reported significant improvements in analyzing code-mixed Roman Urdu and Roman Punjabi, while ? introduced specialized models for Bangla-English code-mixed content. For multilingual applications, cross-lingual models demonstrate strong transfer capabilities. ? successfully employed advanced techniques for zero-shot transfer from English to Hindi, while ? highlighted the effectiveness of cross-lingual transfer learning for financial sentiment analysis in Spanish.

The Resource-Performance Tradeoff

Despite their impressive results, specialized sentiment analysis models present implementation challenges. As ? discusses, there exists a clear tradeoff between performance and efficiency, with advanced models requiring computational resources. Language-specific models typically contain millions of parameters (?), while larger multilingual models can expand to hundreds of millions or even billions of parameters (??).

The implementation of these specialized models requires not only significant computational infrastructure but also technical expertise in model selection, fine-tuning, and deployment. ? demonstrated meaningful improvements after fine-tuning on domain-specific Russian data, but such fine-tuning processes demand substantial ML engineering knowledge. For many social science researchers and practitioners without technical backgrounds, these requirements present barriers to adoption.

Additionally, the literature emphasizes the importance of high-quality, domain-specific training data. ? noted better performance for Nigerian languages with larger available datasets, while ? highlighted the challenges of limited data availability for linguistically diverse languages like Finnish, Hungarian, and Bulgarian. These data requirements further complicate implementation for researchers working with low-resource languages or specialized domains.

The Potential of General-Purpose Foundational LLMs

While specialized models dominate the literature on multilingual sentiment analysis, relatively little attention has been given to the capabilities of general-purpose foundational LLMs without task-specific fine-tuning. This represents a gap in the research, particularly given the findings of ?; ?, and others regarding the unexpected effectiveness of English-prompted models on cross-linguistic tasks.

Recent studies by ? and ? have begun to explore the potential of general-purpose LLMs like Mistral-7B and LLaMA2 for sentiment analysis in Italian and across multiple languages, respectively. However, these studies remain preliminary, and comprehensive evaluations of open-source foundational LLMs across diverse languages remain limited. This gap is particularly notable given the practical advantages general-purpose LLMs might offer. As highlighted by ? in their work on German clinical texts, leveraging foundational models without extensive fine-tuning could reduce implementation barriers. Similarly, ? observed promising results using general instruction-tuned models for sentiment analysis in Russian and English, suggesting the potential of generalist models for cross-linguistic applications.

The limited existing research on general-purpose LLMs for multilingual sentiment analysis reveals several patterns worth investigating. ? found that language-specific prompting strategies significantly impact performance for Lithuanian sentiment analysis, while ? documented varying effectiveness across Germanic and Romance languages depending on prompt design and model architecture.

The phenomenon documented by ?; ?, and others—that English prompting often improves cross-linguistic performance—suggests that foundational LLMs may possess unexpectedly robust multilingual capabilities even without specialized training. This aligns with observations from ? and ?, who found promising results

with general-purpose models in Russian and Italian contexts, respectively. Our study addresses this research gap by evaluating leading open and closed weights foundational LLMs with French texts and their translation.

Data & Methods

This study assesses the efficacy of general-purpose large language models in non-English sentiment analysis through a comprehensive evaluation of open and closed-source models on French text. We evaluate whether foundational models demonstrate sufficient cross-linguistic capabilities for sentiment analysis without task-specific fine-tuning, comparing their performance to traditional easy-to-use dictionary-based methods and examining the impact of prompt language on analytical outcomes.

Data Collection and Preparation

Our corpus consists of 2683 French-language news articles from 13 media sources collected from the Eureka database spanning 1991-2025. It’s an exhaustive collection of every article published about open-source in what Eureka calls its “Major daily newspapers (FR)” category. 13 sources, 9 from Québec, 1 from Ontario, and 3 from France, ensuring a diverse representation of the francophone media landscape. The articles were selected based on the following search query:

```
TEXT= ("logiciel libre" | "logiciels libres" | "open source" | "open-source" | "logiciel  
open source" | "logiciels open source" | "code source ouvert" | "software libre"|  
"free software" | "code source libre" | "Free Software Foundation" | "Richard  
Stallman")
```

The earliest date of January 1st, 1991 was selected to ensure a comprehensive representation of the evolution of open-source discourse over time. 1991 is the year of the first release of the Linux kernel, a pivotal moment in the history of open-source software. January 1st, 2025 was selected as the end date to ensure that the articles were all published before the start of the study. This covers a period of 34 years, allowing us to analyze the evolution of open-source discourse over time. However, the first article collected was published on June 23rd, 1995, as no articles fitting the query were published before that date. Table ?? shows the distribution of articles by source media, with a total of 2683 articles collected from 13 different sources.

Table 1: Summary of Collected News Articles Discussing Open Source Software by Source Media

source_media	article_count	earliest_date	latest_date
Libération	769	1995-06-23	2024-11-04
Monde, Le	666	2001-01-09	2024-12-26
Figaro, Le	402	1998-10-21	2024-12-04
Devoir, Le	318	1998-07-20	2024-10-26
Soleil, Le (Québec, QC)	179	1998-11-08	2023-11-18
Presse, La	152	1998-03-26	2017-10-14
Journal de Montréal, Le	65	2006-12-20	2023-04-24
Journal de Québec, Le	45	2007-03-24	2024-10-19
Tribune, La (Sherbrooke, QC)	43	2000-03-29	2022-12-03
Droit, Le (Ottawa, ON)	17	1998-11-10	2014-07-17
Nouvelliste, Le (Trois-Rivières, QC)	13	2000-01-13	2022-03-05
Quotidien, Le (Saguenay, QC)	10	2008-11-01	2019-04-10
Voix de l’Est, La (Granby, QC)	4	2010-09-17	2015-02-14

We developed a Python-based web scraper using Selenium WebDriver to extract article content more conveniently from the database. Eureka provides a way to download articles in PDF format in batches of 50 articles. Downloading 2683 articles this way is very time-consuming and the PDF format is not very convenient for text analysis. The scraper allowed the easy download of all articles in HTML format, which is more convenient for text analysis. All HTML documents underwent preprocessing with custom R functions

to extract publication metadata (date, source, title) and full-text content. This resulted in a temporally diverse exhaustive corpus suitable for sentiment analysis across multiple methods. The extraction process preserved document structure while standardizing format, enabling consistent comparison across analytical approaches.

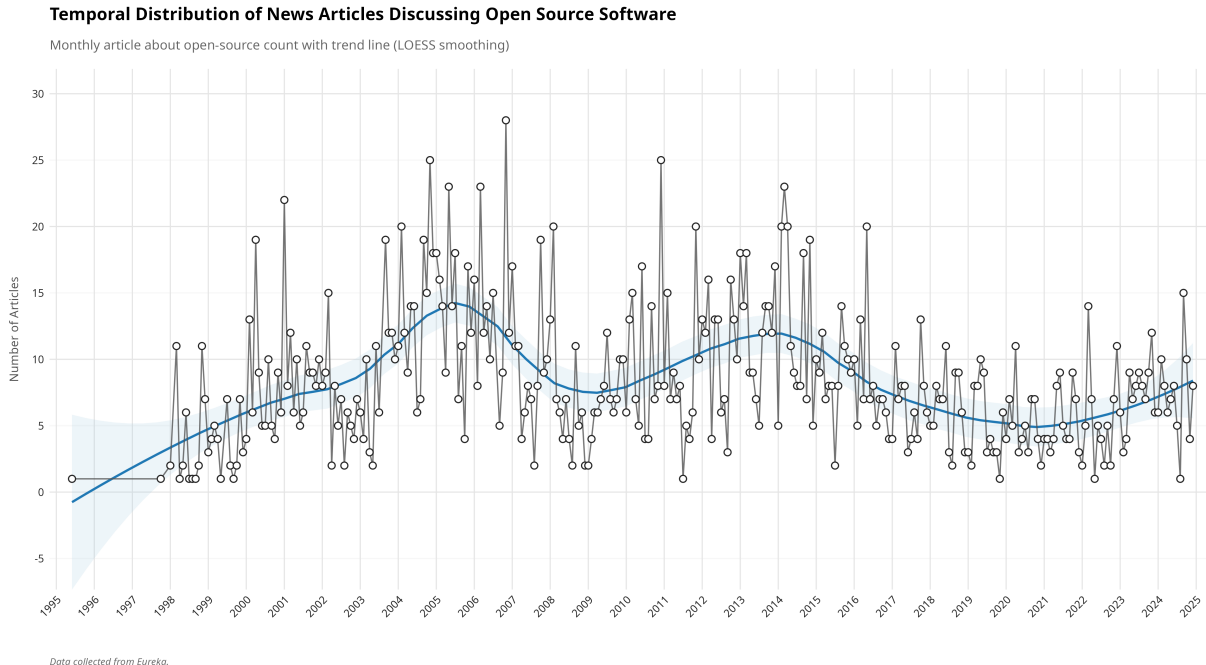


Figure 1: Temporal Distribution of News Articles Discussing Open Source Software

Translation

To facilitate cross-linguistic analysis, we translated the original French text into English using Google Translate. To manage the translation of our large corpus (2683 articles), we developed a systematic approach using R. We batched the French articles into Word documents (.docx) using the `officer` package, with each batch containing approximately 900 articles. Each article was marked with a unique delimiter (###ARTICLE_START###) and its original document ID to maintain traceability. These Word documents were then translated using Google Translate’s document translation service. After translation, we processed the translated documents to extract the English text while preserving the document IDs. The extracted translations were then merged back into our original dataframe, creating parallel French-English text columns for each article.

This approach allowed us to efficiently translate large volumes of text for free while maintaining document integrity and preserving the connection between original French content and its English translation. The process was automated using R scripts, ensuring consistent handling across all documents and enabling systematic comparison of sentiment analysis results across languages.

Manual Annotation

To establish a reliable benchmark for evaluating model performance, we created a validation dataset consisting of 200 randomly selected sentences from our corpus.

We first selected 200 random sentences from across all articles in the corpus. Each sentence was then translated into English using Google Translate and the `polyglotR` package to facilitate cross-linguistic analysis. The 200 sentences were manually annotated using a scale ranging from -1 to 1, where -1 represents very negative sentiment, 0 indicates neutral content, and 1 signifies very positive sentiment.

To streamline the annotation process, we developed a custom Shiny application that provided an intuitive interface for human coders. The application featured keyboard shortcuts for rapid scoring and included visual reminders of the sentiment scale definitions. This approach ensured consistent annotation practices and significantly improved the efficiency of the manual coding process. The resulting annotated dataset served as our ground truth for evaluating the performance of both dictionary-based methods and LLM approaches in sentiment analysis tasks.

A major limitation of this project is that the manual annotation was done by a single coder with a small sample of sentences. This limits the reliability of the ground truth dataset, as the inter-annotator agreement was not measured. This should be considered when interpreting the results of the study.

Dictionary-Based Sentiment Analysis

To establish classical baseline sentiment measures, we employed lexicon-based methods on both the original French text and its English translation. We used a two-phase approach to ensure comprehensive analysis across languages.

For the French corpus, we implemented a customized version of the Lexicoder Sentiment Dictionary (LSD) methodology (?). First, we segmented all articles into individual sentences, maintaining document structure through unique sentence identifiers. Each sentence underwent preprocessing: conversion to lowercase, removal of punctuation, and elimination of French stopwords using the `quanteda` package. We then applied the French Lexicoder Sentiment Dictionary (`frlsd`), a specialized lexicon adapted for French text analysis, to identify positive and negative sentiment terms in each sentence. Sentiment scores were calculated as a normalized tone index representing the ratio between positive and negative sentiment markers within each text segment, with the formula: $(\text{positive_terms} - \text{negative_terms}) / \text{total_words}$.

For the English translations, we implemented a more sophisticated preprocessing pipeline based on the standard Lexicoder Sentiment Dictionary methodology (?). Beyond basic cleanup, we applied specialized functions to handle contractions, preserve proper nouns, and process negation expressions (e.g., “not good” recognized as negative rather than positive). The English LSD includes dedicated dictionaries for negated terms, allowing us to account for the sentiment-flipping effect of negation (e.g., “not happy” is properly categorized as negative sentiment). The tone index was calculated as: $((\text{positive} + \text{negated_negative}) - (\text{negative} + \text{negated_positive})) / \text{total_words}$, providing a more nuanced assessment of sentiment.

This dual-dictionary approach provided a methodologically sound baseline against which to evaluate LLM performance while enabling the assessment of potential information shifts during translation.

LLM-Based Sentiment Analysis

Our LLM evaluation framework incorporated an extensive assessment of multiple general-purpose foundational models, carefully selected to represent the current landscape of both open and closed-weight architectures. We conducted a total of 19,800 individual prompts across 11 different models to thoroughly evaluate their sentiment analysis capabilities in cross-linguistic contexts. This scale of evaluation allowed us to draw conclusions about model performance across language boundaries and prompt conditions.

Our model selection reflects a deliberate effort to represent the current landscape of LLMs across different architectures, parameter scales, and accessibility levels. For our open-weight model selection, we incorporated a diverse range of architectures and parameter sizes to ensure a comprehensive representation of current capabilities. The smallest model in our evaluation was Llama 3.2 (1B), accessed through the Groq API service, which allowed us to establish a baseline for minimal parameter count performance. We also included Llama 3.2 (3B), accessed via Fireworks.ai, which represents a modest increase in parameter size while maintaining computational efficiency. Gemma 2 (9B), Google’s mid-sized open model accessed via Groq, provided an intermediate step in model scale. Models of these sizes are able to be run on consumer-grade hardware, making them accessible for researchers without access to large computational resources. Mistral Saba (24B), also accessed through Groq, represented a substantial increase in parameter count with Mistral AI’s specialized architecture. QWQ (32B), developed by Alibaba Cloud and accessed via Fireworks.ai, introduced another architectural approach with a larger parameter count. The second largest open-weight model in our evaluation was Llama 3.3 (70B), Meta AI’s advanced offering accessed through Fireworks.ai. We

also included DeepSeek R1 Basic (671B) via Fireworks.ai, the largest open-weight model in our evaluation, which provided a unique opportunity to assess the performance of a model with an exceptionally high parameter count.

The closed-weight models in our evaluation were selected to represent leading commercial offerings with established performance records. These included Claude 3.5 Haiku, Anthropic’s efficient conversational model recognized for its instruction-following capabilities; Gemini 2.0 Flash, Google’s streamlined but powerful model with enhanced multilingual support; DeepSeek Chat, a specialized conversational variant from DeepSeek AI; and GPT-4o, OpenAI’s advanced multimodal model renowned for its cross-linguistic proficiency. This combination of open and closed-weight models created a comprehensive evaluation framework spanning different architectural approaches, training methodologies, and parameter scales. We selected these models based on their prominence in the field and their affordability, as they are all available for free or at a low cost compared to some of the leading models on the market which are often prohibitively expensive.

To systematically evaluate each model’s performance across different language contexts, we implemented a rigorous experimental design consisting of three distinct analytical conditions. The first condition [Fr -> Fr] involved analyzing the original French text with French prompts, which assessed each model’s native language understanding when both input text and instructions were in the same non-English language. The second condition [En -> Fr] evaluated French text with English prompts, directly examining the cross-linguistic prompting effect documented in previous research. The third condition [En -> En] used English translations of the original French sentences with English prompts, providing insight into how translation might impact sentiment analysis performance. This three-condition approach enabled us to isolate the effects of both prompt language and text language, allowing for a detailed analysis of how each factor impacts model performance and measurement of potential information loss during translation.

All prompts across both languages followed a meticulously standardized structure to ensure valid cross-linguistic comparisons. Each prompt began with a clear task definition explicitly requesting sentiment analysis of the provided text. We included a comprehensive sentiment scale documentation that defined five distinct anchor points: -1.0 (strong negative sentiment, described as highly critical, hostile, or pessimistic content), -0.5 (moderate negative sentiment, described as somewhat negative, disapproving, or concerned content), 0.0 (neutral sentiment, described as factual, balanced, or neither positive nor negative content), 0.5 (moderate positive sentiment, described as somewhat positive, approving, or optimistic content), and 1.0 (strong positive sentiment, described as highly supportive, enthusiastic, or optimistic content). The prompts explicitly permitted the use of intermediate values between these anchor points to allow for fine-grained sentiment assessment. We included cultural context guidance instructing models to consider linguistic and cultural nuances specific to the language being analyzed. Finally, all prompts contained strict response format constraints requiring only numerical values without any explanations or additional text, ensuring consistent and comparable outputs across all models. French prompts were translations of the English prompts, maintaining identical structure, meaning, and nuance to ensure valid cross-linguistic comparison.

Here is the prompt used for the [En -> Fr] condition: ³

```
en_prompt_fr_text <- function(text) {
  paste0("Please analyze the sentiment of the following French text and provide a single
  numerical rating according to this scale:
  Sentiment Scale:
  -1.0: Strong negative sentiment - highly critical, hostile, or pessimistic content
  -0.5: Moderate negative sentiment - somewhat negative, disapproving, or concerned
  content
  0.0: Neutral sentiment - factual, balanced, or neither positive nor negative content
  0.5: Moderate positive sentiment - somewhat positive, approving, or optimistic content
  1.0: Strong positive sentiment - highly supportive, enthusiastic, or optimistic content
  You may also use values between these points for intermediate sentiment levels")
}
```

³The prompt used for the [Fr -> Fr] condition is identical, except that the text is in French and the prompt is in French. See the GitHub repository in `src/93_prompts.R` for the exact prompt used.